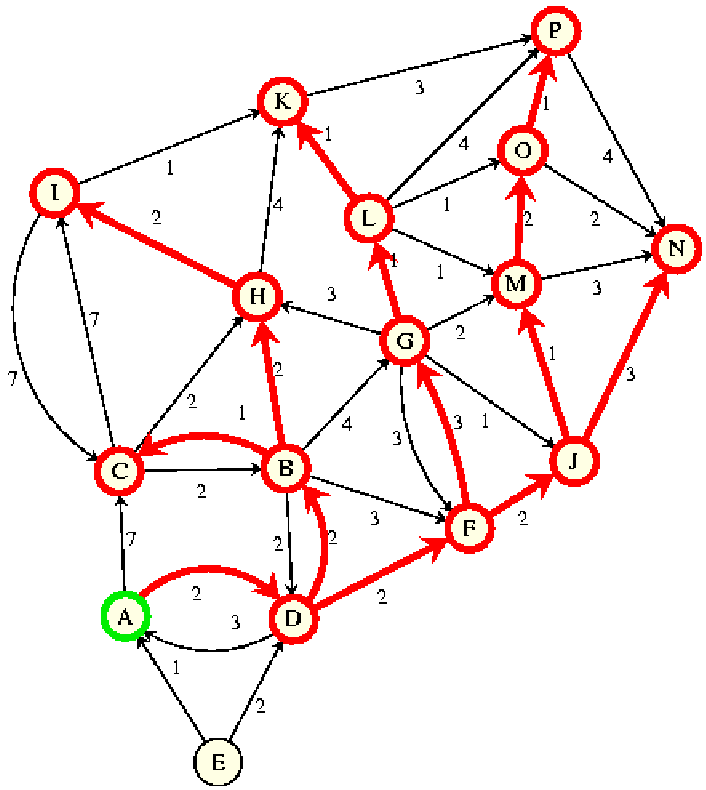
1. What is a computer network?
2. Discuss various types of networks topologies in computer network.
3. Also, discuss advantages and disadvantages of each topology.
4. What are the applications of Computer Networks?
5. What is OSI Model? Explain the functions, protocols and services of each layer?
6. Explain the following:-
    I. LAN
    II. MAN
    III. WAN
7. What are the network components? What do you mean by client-server network?
8. Write some protocol names of each layer of TCP/IP model.
9. Explain protocol, layer, peer and interface using layered diagram.
10. Explain connection multiplexing, connectionless and connection-oriented services.
11. Explain OSI and TCP/IP model
12. Describe HDLC frame format.
13. How many types of HDLC frame there are? Explain control field formation of each of those.
14. When and why we use point –to-point connection?
15. Draw the ppp frame format. Explain each field.
16. Explain the diagram for bring a line up and down in a PPP communication.
17. ATM has how many cell formats? Explain the commonly used one.
18. Why Standardization is necessary? Explain with example
19. How does CSMA/CD work? Explain
20. What do mean by token? Explain IEEE 802.4 and 802.5.
21. Why not just one big LAN?
22. What devices are available for LAN's interconnection. Mention which layer they work.
23. Explain HUB and Bridge. Compare them.
24. Give a bridge learning example.
25. What will happen if there is a loop between two nodes?
26. What is spanning tree? How it helps bridge.

27. Compare and contrast between routers and bridge.
28. What do you mean by cut-through switching?
29. Compare and contrast between switch and bridge.
30. What is FDDI? How does it work?
31. What are the advantages of FDDI?
32. Draw the FDDI frame formation.
33. Differentiate between Ethernet and token ring.
34. If there is a network like 172.168.0.0/17, then determine the followings:
    a. How many subnets are possible?
    b. How many hosts are there?
    c. What is the subnet mask of the network?
    d. In the 1st subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?
    e. In the 18st subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?
35. What are the physical media available for Ethernet, Fast Ethernet and Gigabit Ethernet? Also note their maximum segment length.
36. **What leads to fast Ethernet?
37. What are the Ethernet and fast Ethernet and gigabit physical media types? White down their maximum-segment-length.
38. Explain Non-adaptive & adaptive routing algorithm.
39. What do you mean by the optimality principle and sink tree?
40. What do you mean by routing metrics? Explain with figure and example.

41. 

Use shortest path algorithm to find shortest path from A to M.

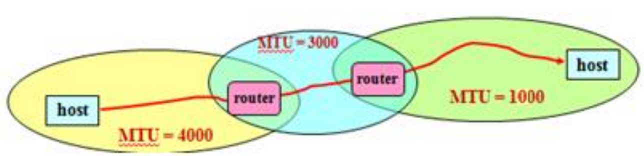
42. State the steps of the shortest path algorithm.

43. Explain with figure the count to infinity problem in DVR algorithm.

44. Briefly explain the steps of link state routing.

45. Why we need interconnecting networks and how to do that? How the available networks differ from each other?

46. Why we need packet fragmentation? Do fragmentation for the following network



47. What are the fragmentation related fields? Explain.

WEEK-5

48. What are the network layer protocols? What are purposes of using an IP address?

49. Why deliveries of a packet need two level of addressing e.g. IP and MAC?

50. Explain static and dynamic mapping. State their limitations.

51. Show with figure that ARP request is broadcast and reply is unicast.
52. Show ARP packet formation and encapsulation. What are the 4 cases of using ARP.
53. A host with IP address 130.23.3.20 and physical address B23455102210 has a packet to send to another host with IP address 130.23.43.25 and physical address A46EF45983AB. The two hosts are on the same Ethernet network. Show the ARP request and reply packets encapsulated in Ethernet frames.
54. Show RARP packet formation and encapsulation.
55. Explain using figure the four cases of using ARP.
56. What is the purpose of DHCP protocol and how does it work?
57. Draw state-transition diagram for DHCP. Explain all six states.
58. Show the DHCP message exchange between client-server.
59. Draw IPv4 and IPv6 datagram format and explain about its each field briefly.
60. Define MTU and the the fields of fragmentation with appropriate figure.
61. What are the merits of IPv6 over IPv4?
62. Explain three transition strategies from IPv4 to IPv6 .
63. Explain the fragmentation procedure of IPv6 packet and compare it to IPv4 fragmentation.
64. Explain Home address and care-of address, Home agent and foreign agent with appropriate figure.
65. Explain multiplexing and de-multiplexing between network and trasport layer using appropriate figure.
66. To communicate with a remote host, a mobile host goes through three phases: agent discovery, registration, and data transfer. Time diagram.
67. Communication involving mobile IP can be inefficient. Explain the scenarios.

Week-6

68. Differentiate between ATM cell and TCP packet.
69. Describe both the ATM cell types with appropriate cell formation figure.
70. Write short note on ATM network and transport layer.
71. How does TCP/UDP (transport layer protocol) works?

72. Why would anyone use UDP? Clarify your answer with some applications where the UDP is used.
73. What is the difference between UDP and TCP internet protocols?
74. What are the Challenges of Reliable Data Transfer?
75. How TCP supports for TCP Support for Reliable Delivery?
76. What is ISN? How does it work? Explain using TCP packet transfer.
77. How to establish a TCP connection? What is purpose of ISN is establishing the TCP connection?
78. Use the TCP header to the TCP connection setup sequentially.
79. What happens when a SYN pkt gets lost?
80. What are the Reasons for Retransmission? Use figure. What do mean by RTT and what is the use of it?
81. What were the motivations for TCP sliding window? How does it work?
82. Explain how timeout base retransmission is inefficient.
83. How to close a TCP connection?

Week-7

84. Write short notes on UDP and RPC.
85. What is purpose of RTP?
86. Show the position of RTP in the protocol stack, Packet nesting and RTP header formation with explanation.
87. What are the basic functions of TCP?
88. Draw the TCP segment header and explain the task of important fields.
89. Explain how TCP manages a connection in different cases?
90. Explain with flow diagram TCP connection management finite state machine.
91. What are the TCP packet retransmissions policies?
92. What are solutions to Silly Window Syndrome?
93. How TCP controls the congession?
94. What do you mean by slow start algorithm?
95. HOw many timers are there in TCP protocol?

96. Write short note on all those timers. (Help from book)

Week-8

97. What is the purpose of ATM adaptation layer? Describe all proposed AAL protocols for different traffic types.

98. Draw the detailed view of AAL.

99. What do mean by a flying LAN?

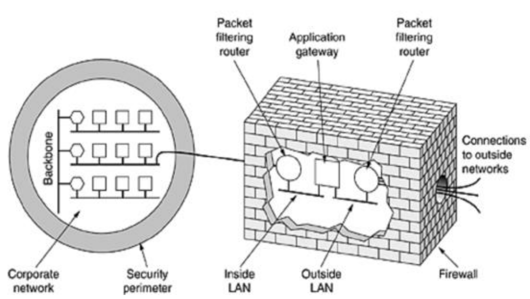
100. Write short note on: (take help from book, only figure and basic purpose in 5-6 lines is enough)

    a. The Electromagnetic Spectrum
    b. Radio Transmission
    c. Microwave Transmission
    d. Infrared Transmission
    e. Light Transmission

101. Explain each of the following firewall types:

    a. Packet filtering
    b. Circuit gateways
    c. Application gateways

Figure 8-29. A firewall consisting of two packet filters and an application gateway.



102. Explain the above firewall setup.

103. Suppose we want to allow inbound mail (SMTP, port 25) but only to our gateway machine. Also suppose that mail from some particular site SPIGOT is to be blocked. Design the firewall.

104. Now suppose that we want to implement the policy “any inside host can send mail to the outside”. Design.

105. Write short notes on DoS and DDoS.
106. Differentiate between wired and wireless TCP.
107. How to measure the network performance.
108. How to do System Design for Better Performance.
109. Write short note on Fast TPDU Processing.
110. How to Optimize the high speed networks.
111. Explain the characteristics of gigabit network and design issues for this network.

Week – 9

112. Differentiate between 1G, 2G, 3G and 4G.
113. Write short notes on IMPS and AMPS.
114. How AMPS accumulate more calls than IMPS? PP 154-Book
115. Step by step explain MOC with flow diagram.
116. Draw the basic diagram of GSM system. Show the channel distribution and framing structure of GSM network.
117. What are the improvements in IMT-2000 network?
118. How to handle a call during cell change. Explain soft/ hard handoff with figure.
119. How to achieve 100 Mbps capacity. Take help from the book.
120. What are 3 choices of fast-Ethernet?
121. Write short notes on: 100 BASE T4, 100 Base TX, 100 Base FX
122. Write short notes on:
    a. 1000 BASE SX *fiber - short wavelength*
    b. 1000 BASE LX *fiber - long wavelength*
    c. 1000 BASE CX *copper - shielded twisted pair*
    d. 1000 BASE T *copper - unshielded twisted pair*
    e. 8B/10B Encoder
    f. A buffered distributor

123. Explain with figure the task of MII and GMII.

Week - 10

124. Define DNS , domain name and sub domain names.
125. Explain the dns naming structure. Give some examples of TDLs and CCDs
126. <http://jad.shahroodut.ac.ir/article_147_44.html > what does each part means?
127. What do you mean by The Resource Record? Give some examples.
128. Differentiate between Iterative vs Recursive Name Servers
129. How DNS server work for web and mail?
130. Show Example of a resolver looking up a remote name in 10 steps using flow diagram.
131. Draw the block diagram/ architecture of email system and explain the services
132. Show the differences between paper mail and electronic mail.
133. What are the fields used in the RFC 5322 message header and what are fields added by MIME. Explain each field.
134. Draw all the possible scenarios of e-mail architecture
135. Explain push and pull messages.
136. What are the services provided by a user agent and message transfer agent?
137. Show the connection establishment and message transfer and connection termination between MTA client and MTA server.
138. What is MAA?
139. Write short note on POP3 and IMAP4.
140. What protocol destination user uses to communicate with mail server and get the messages?
141. Write short note on MIME, Base64 encoder.

Week – 11

142. Write short note on SNMP, MIBS, TRAP. What is the purpose of MIBS?

143. Show SNMP manager and an SNMP agent communicate using the SNMP protocol, also explain the meaning of each massage

144. Explain the security level in different version of SNMP.

145. How the HTTP request and reply works?

146. The Request Method can be:

    a. GET, HEAD, DELETE, PUT, POST, TRACE, OPTIONS

    Explain all these methods. Try with example.

147. What are the tasks of header lines in http protocol? Explain with example.

148. Give Example of a HTTP response and its meaning.

149. Differentiate between multiple connection and persistent HTTP connection.

150. Difference between HTTP 1.0 and HTTP 1.1

Week - 12

151. What is world wide web? Explain the client server scenario of WWW.

152. What do you mean by

    a. **The Hypertext concept**
    b. **The Hypermedia concept**
    c. **WEB Browser**
    d. **WEB Server**
    e. **Uniform Resource Locator (URL)**

153. Find the similarities and dissimilarities between internet and WWW.

154. Named some people who cause security problems and why?

155. Draw the general encryption model for symmetric key cipher.

156. Explain Substitution Ciphers. How many type of Substitution Ciphers are there. Explain each.

157. Explain how to do Transposition Ciphers. Use SOHEL as key to cipher the following text < Facts of the day for political enthusiasts: Vietnam's first reaction to Chinese provocation - a Vietnamese patrol boat exchanged water canon fire for over one hour with 15 Chinese boats >

158. What do you mean by product cipher? Write short note on P-Box and S-box.
159. How does DES work? Show and explain using diagram.
160. Explain the triple encryption using DES. Why the encryption and decryption follows EDE and DED respectively?
161. Write in short how AES works. Draw its simple block digram.
162. Explain Cipher block chaining. (a) Encryption. (b) Decryption. What are the benefits of block chaining cipher?
163. How does RSA work. Use some own example to explain the procedure.
164. What is DS (digital signature). Differentiate it with general signature.
165. Explain Symmetric-key Signatures and its problem.
166. Explain public-key cryptography using block diagram.
167. What do you mean by Message Digest (MD).
168. What are the Message Digest properties?
169. Show Digital signatures using message digests.
170. How to use of SHA-1 and RSA for signing nonsecret messages.

Week – 13

171. What forces us to go for IPsec?
172. Why IPsec is algorithm independent?
173. The IPsec authentication header in transport mode for IPv4.?
174. Explain each field of above AH header.
175. Show (a) ESP in transport mode (b) ESP in tunnel mode.
176. Explain four-packet handshake. What is the purpose of it?
177. What is the purpose of Authentication Protocols?
178. Show and explain Two-way authentication using a challenge-response protocol (shared secret key)
179. Show the shortened two-way authentication protocol.
180. Explain reflection attack.
181. What do you mean by HMAC. What are the things we need to build HMAC.
182. Show and explain the authentication procedure using HMAC.

183. Explain “The Diffie-Hellman Key Exchange”.

184. Give an example of “The Diffie-Hellman Key Exchange”.

185. What do you mean by “The man-in-the-middle attack”

186. Use a n example to show how “The man-in-the-middle attack” is done.

187. Describe the AH procedure using KDC.

188. What is purpose of session key here?

Good Luck